

SUMMARY OF QUALIFICATIONS

- Computer Science student (Faculty of Computers and Artificial Intelligence, Cairo University) with a focus on systems architecture, networking, and cybersecurity.
- Google Cybersecurity Professional Certificate and Cisco Introduction to Cybersecurity holder, with hands-on exposure to Linux, packet analysis, intrusion detection, and SIEM tooling.
- Full-stack builder across C++, C#, Java, Python, and JavaScript, delivering role-based, database-backed applications from POS systems to desktop finance tools.
- Strong grounding in OOP, data structures, system design, and MVC/MVVM architecture, paired with entrepreneurship training from the TIEC program.

EDUCATION

Faculty of Computers and Artificial Intelligence, Cairo University (FCAI-CU) | *Cairo, Egypt*

Bachelor of Science in Computer Science — Junior Year

CERTIFICATIONS

- Google Cybersecurity Professional Certificate — Coursera, July 2026 — [Verify ↗]
- Introduction to Cybersecurity — Cisco Networking Academy / ELCC, July 2026 — [Verify ↗]
- Digital Safety and Security Awareness — Cisco Networking Academy / ELCC, July 2026 — [Verify ↗]
- English for IT: Advice and Time — Cisco Networking Academy / ELCC (OpenEDG), July 2026 — [Verify ↗]
- TIEC Entrepreneurship Program — Creativa Innovation Hub.

TECHNICAL SKILLS

Programming Languages: C++, C#, Python, Java, JavaScript, SQL, HTML/CSS

Databases: SQL Server, SQLite, Relational Database Schemas

Networking & Cybersecurity: Wireshark, Nmap, SIEM Tools, Intrusion Detection Systems (IDS), Cisco Networking Foundations

Concepts & Architecture: OOP, Data Structures, System Design, MVC / MVVM Patterns

Tools & OS: Linux, Git, Zsh, Alacritty Terminal

KEY PROJECTS

Bakery POS System | *Tauri (Rust), React/TypeScript, Tailwind, SQLite*

- Engineered a lightweight, cross-platform desktop POS application using a Tauri (Rust) backend with a React + TypeScript frontend styled in Tailwind CSS.
- Configured local SQLite persistence via tauri-plugin-sql with a full CRUD data-access layer for products, orders, and sales history.
- Built a real-time cashier terminal (cart calculations, discounts, customer/delivery metadata), direct receipt-printer integration, and an Admin Dashboard for revenue and stock-value reporting.

Timeless Wisdom Library | *Django, JavaScript, SQLite, HTML/CSS*

- Engineered a full-stack digital library platform featuring secure role-based access control (RBAC) via Django REST Framework and JWT authentication.
- Implemented catalog browsing, category filtering, and real-time book availability tracking backed by optimized search and relational database queries.
- Designed an intuitive client interface and an administrative dashboard for full lifecycle collection management.

Masroofy — Expense & Finance Management App | *Java, SQLite*

- Co-developed a desktop personal finance management application using Java, structured on MVC/MVVM design patterns for clean UI/Logic separation.
- Architected a local SQLite relational database schema for tracking transactions, expense categories, and budget logs with strict data integrity constraints.
- Authored comprehensive Software Requirements Specifications (SRS) and PlantUML system architecture diagrams to guide modular software development.

Smart Corporate Workspace & Resource Hub | *Python (Flask), SQL Server, HTML/CSS/JS*

- Built a full-stack coworking-space management system on SQL Server with a Flask REST API (14 endpoints) and a dependency-free HTML/CSS/JS frontend.
- Managed the full lifecycle of members, workspaces, reservations, and equipment across multiple hub locations.
- Delivered 6 analytical reports (popular workspace types, inactive members/hubs, equipment usage, member hours) directly against the database.

Advanced Audio Player | *C++, JUCE Framework*

- Engineered a cross-platform desktop audio player using C++ and the JUCE framework for real-time digital audio signal processing (DSP).
- Implemented low-latency audio buffering, playback controls, and dynamic volume/frequency filtering without audio distortion or buffer underruns.
- Designed a responsive custom GUI utilizing JUCE's component hierarchy and event-driven architecture.

Board Game Engine | *C++, CMake, Doxygen*

- Architected a modular, scalable C++ board game engine leveraging Object-Oriented principles (Polymorphism, Abstraction) to decouple core game loops from specific rule sets.
- Implemented a flexible framework supporting 8+ distinct turn-based game variants (e.g., PyramidXO, Infinity, Connect4, Numerical Tic-Tac-Toe, Memory, SUS, and obstacle-based games) via extensible base classes.
- Configured automated C++ build systems using CMake and generated comprehensive API documentation with Doxygen.

Baby Photoshop — Image Manipulation Tool | *C++*

- Developed a high-performance C++ CLI image manipulation tool performing direct pixel-level matrix operations and algorithmic transformations.
- Implemented custom image filters (e.g., edge detection, blurring, color-space conversion, resizing, and rotation) directly handling raw image buffers.
- Optimized algorithm execution and memory usage to ensure efficient processing of large bitmap/image data structures.